Chirayu Nilesh Chaudhari

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Summary

Third-year AI & Data Science student (CGPA 9.29) with three peer-reviewed publications and ICPC Regionalist standing. Work spans **deep learning research**, **RF & signal processing**, and **production ML deployment**.

Skills

Languages: Python, JavaScript/TypeScript, Java
ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, LangChain, spaCy, Stable-Baselines3, Optuna, OpenCV
Domains: Deep Learning, Reinforcement Learning, NLP, Computer Vision, Signal Processing, Data Structures & Algorithms
Systems & Tools: Docker, GCP, Git, Linux, Node.js, React/Next.js, Flask, IoT & Embedded Systems, SDR

Education

B.Tech, Artificial Intelligence & Data Science | CGPA: 9.29/10 Aug 2024 – May 2028
Amrita Vishwa Vidyapeetham, Bengaluru, Karnataka

Publications & Honors

Journal:

- **Chaudhari, C. N.**, et al. "Thermoelectric Energy Optimization with Deep Reinforcement Learning." Accepted for publication, *Springer Nature* journal, 2025.

Conference:

- **Chaudhari, C. N.**, et al. "Contour based tracking vs Hough transform based tracking: A RC Car experiment." *3rd IEEE World Conf. on Communication & Computing (WCONF)*, 2025.
- **Chaudhari, C. N.**, et al. "The Effects and Classification of Psychoactive Substances: A Machine Learning Approach." *IEEE Intl. Conf. on Computational Systems & Information Technology for Sustainable Solutions (CSITSS)*, 2025.

Achievements:

- **ICPC Regionalist (2025).** Qualified for the International Collegiate Programming Contest Regionals.
- **National Finalist, Top 6, Novo Nordisk Hackathon (2025).** Nationwide healthcare innovation hackathon.

Experience

AI Engineering Intern Dec 2025 – Mar 2026
ApplyBeeAI, Bengaluru, Karnataka

- Built NLP and agentic pipelines with LangChain, spaCy, and Gemini, scaling automated processing across 7+ applicant tracking systems including Naukri and Workday.
- Containerized core AI modules with Docker and deployed them to GCP, working with the founders on early-stage architecture decisions.

Software Engineering Intern May 2025 – Jul 2025
SigNoz, Bengaluru, Karnataka

- Architected a Model Context Protocol (MCP) server from scratch in Node.js and TypeScript, exposing real-time SigNoz trace and log data to AI clients.
- Shipped compatibility across Cursor, Windsurf, and Claude, benchmarking the design against Honeycomb's MCP.

Projects

Expression-Robust 3D Face Reconstruction (DECA) Apr 2026 – May 2026

- Extended the SIGGRAPH 2021 DECA model with a residual correction head over its frozen encoder, fine-tuned on FaceScape to target hard-case expressions.
- Built the CUDA training pipeline and an evaluation harness with per-sample failure diagnostics. The fine-tuned model beat the pretrained baseline.

Agentic Spam Call & Misinformation Detection Jan 2026 – Mar 2026

- Fused Whisper ASR transcription, a Log-Mel spectrogram CNN, and an NLP scam scoring model through weighted late fusion.
- Added an LLM explainability agent that justifies every flagged call, making classifier scores auditable.

Suspicious Signal Detection using ML & SDR Nov 2025 – Dec 2025

- Captured live RF I/Q in ISM bands with an ADALM-PLUTO SDR and wrote a synthetic generator injecting parameterized QPSK bursts.
- Benchmarked SVM-RBF and Random Forest metadata classifiers against a custom PyTorch 1D CNN on raw I/Q, which caught temporal and spectral anomalies the others missed.